

EMILY DEEB

Spatial Data Scientist | Urban Analytics | MSc Urban Spatial Science, CASA UCL
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PROFESSIONAL SUMMARY

Spatial data scientist and urban analyst with five years of professional practice delivering city-scale GIS, geospatial analysis, and statistical modelling, now completing an MSc in Urban Spatial Science at CASA UCL. Industry dissertation conducted with CACI Ltd applying Python-based machine learning, predictive modelling, spatial autocorrelation, and data analysis to national retail spending and urban morphology datasets. Experienced in building reproducible data pipelines, stakeholder communication, and translating complex spatial findings into evidence-based decisions. Seeking a role in spatial data science, geospatial analysis, urban analytics, or location intelligence.

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy, Matplotlib, GeoPandas, Scikit-learn), R (RStudio), SQL, React.

Data Science & ML: Classification, Clustering, Regression, Statistical Analysis, SHAP, Spatial Autocorrelation, Pre-modelling Diagnostics, Predictive Modelling

Data & Viz: Data Pipelines, Reproducible Workflows, Data Visualisation, Power BI, Quarto, Jupyter, Git

Spatial & GIS: QGIS, ArcGIS, GeoPandas, Spatial Network Analysis, Spatial Data Science, Urban Morphology, Google Earth Engine API

Design & Other: AutoCAD, Rhinoceros, Adobe Suite (Photoshop, Illustrator, InDesign), Microsoft Office, UI/UX (CalArts certified)

SELECTED PROJECTS

For full portfolio: <https://emilydeeb.com/>

Retail Spending & Urban Morphology Prediction | 2026 Industry Dissertation — CACI Ltd x CASA UCL GeoDS Programme

- Investigated whether urban morphological fabric type predicts retail spending patterns across UK cities, independently of demographic composition, using CACI proprietary consumer spending datasets and HiMoC morphometric classifications.
- Built Python-based machine learning pipeline using GeoPandas, scikit-learn, SHAP, and spatial autocorrelation analysis; applied pre-modelling diagnostics and statistical modelling to produce reproducible, documented analytical outputs.
- Communicated findings through data visualisation and written reporting to CACI's Consumer and Location Intelligence team; presented at GeoDS Data Partner Forum, Saïd Business School, Oxford.

Food Shock — Global Food Supply Crisis Tracker | 2026 CASA UCL Group Project — React, Leaflet, FAO Data, Python

- Built an interactive web application visualising global food supply chain disruptions using FAO trade flow data (200 countries, 82 commodities), commodity price indices, and spatial mapping.

London Crime Density & Urban Amenities Analysis | 2026 CASA UCL — Python, Machine Learning, Spatial Analysis, London Datastore

- Analysed 1.78 million Metropolitan Police Service crime records across 679 London wards to investigate relationships between urban amenity distribution and crime density.
- Applied multiple regression, K-Nearest Neighbours, and Random Forest models with SHAP decomposition; found nightlife amenities strongest predictors of person/society crime, retail nodes driving property crime.

PROFESSIONAL EXPERIENCE

Buro Happold, London | June 2026 Consultancy Project — Building a Better Future Scheme, UCL Bartlett

- Selected through a competitive UCL application to deliver a two-week industry consultancy project on smart campus digital strategy with Buro Happold's Digital Advisory team.

Derman Verbakel Architecture, Tel Aviv | 2020 – 2025 Team Leader — Urban Design & GIS

- Led a team of 4 architects across 15 concurrent projects spanning block to city scale for municipal and private clients including Eilat Municipality and the Israeli Authority for Urban Renewal.
- Developed and led geo-computational workflows in QGIS and ArcGIS for morphological mapping, site feasibility analysis, market analysis, and statistical profitability modelling — improving office efficiency by 30%.
- Independently collected, cleaned, and aggregated spatial datasets from scratch for Sakhnin city centre regeneration; delivered evidence-based spatial reports and cartographic visualisations to municipal planning committees.
- Trained 15+ team members in GIS tools; developed reproducible, documented spatial workflows adopted across the office.
- Key projects: Transportation Masterplan of Eilat; Regeneration of Sakhnin City Centre (1,200 units); TOD Regeneration Plan Rishon LeTsiyon (3,000 units); City expansion masterplan (100,000 units, confidential).

SYL Studio Yigal Levi Architecture, Jerusalem | 2019 – 2020 Architect Intern

- Gained foundational experience in architectural design, documentation, and site coordination.

EDUCATION

CASA, University College London | 2025 – 2026 MSc Urban Spatial Science

- Specialising in computational urban analysis, spatial data science, geocomputation, statistical modelling, and machine learning for cities.
- Modules: GIS & Science, Quantitative Methods, Data Science for Spatial Systems, Building Spatial Apps with Big Data, Urban Data Visualisation, Urban Simulation, Dissertation.

Bezalel Academy of Arts and Design | Jerusalem, 2015 – 2020 B.Arch Architecture

- Graduation Project: Destination: Jericho — research-led urban design study using QGIS, spatial analysis, and data visualisation. Nominated for David Azrieli Architecture Prize.

LANGUAGES & CERTIFICATIONS

Arabic (native), English (C1 — fluent), Hebrew (fluent)

Licensed Architect — Israeli Labor Department, 2025, UI/UX Design — CalArts certified

INTERESTS

Cooking, drawing, exploring cities on foot, collecting fun facts, and always learning something new.